1 July 31, 2025 (Class 1)

Online learning is a sequential learning problem

- No concept of training/test data. Data set is not given at the start at once.
- Decisions on the go. No concept of training loss/accuracy.

Algorithm 1 Online Learning Framework

for $t = 1, 2, \ldots$ **do**
  ALG selects action $a_t \in \mathcal{A}$
  Environment/adversary selects outcome $y_t \in \mathcal{Y}$ and/or loss function $l_t \in \mathcal{L}$
  ALG incurs loss $l(a_t, y_t)$
end for

1.1 Learning with Expert Advice

Algorithm 2: Learning with Expert Advice Framework
input: N experts
for $t = 1, 2, \ldots$ **do**
- Each expert provide advice: $f_{i,t} \in \mathcal{D}$
- ALG selects action $a_t \in \mathcal{D}$
- Environment/adversary selects an action $y_t \in \mathcal{Y}$
- ALG incurs loss $l(a_t, y_t)$

end for

1.2 Weighted Majority Algorithm

Algorithm Weighted Majority Algorithm
Input: N experts, learning rate $\alpha \in [0, 1)$
Initialize: $w_{i,1} = 1 \ \forall i \in [N]$
for $t = 1, 2, \ldots$ **do**
- Input: Expert recommendations $f_{i,t} \in \{0, 1\}$
- Predict: $p_t = \mathbb{I}\{\sum_{i:f_{i,t}=1} w_{i,t} \geq \sum_{i:f_{i,t}=0} w_{i,t}\}$
- Observe: $y_t \in \{0, 1\}$
- Update weights: $w_{i,t+1} \leftarrow w_{i,t} \cdot \alpha^{\mathbb{I}\{f_{i,t} \neq y_t\}}$

end for

We define the potential function as

$$\Phi(t) = \sum_{i=1}^{N} w_{i,t-1}$$

1.2.1 Mistake bound of Weighted Majority Algorithm

We have,

$$\begin{aligned}
\Phi_{t+1} &= \sum_{i:f_{i,t}\neq y_t} w_{i,t} + \sum_{i:f_{i,t}=y_t} w_{i,t} \\
&= \sum_{i:f_{i,t}\neq y_t} \alpha w_{i,t-1} + \sum_{i:f_{i,t}=y_t} w_{i,t-1} \\
&= \alpha\left(\Phi_t - \sum_{i:f_{i,t}=y_t} w_{i,t-1}\right) + \sum_{i:f_{i,t}=y_t} w_{i,t-1} \\
&= \alpha\Phi_t + (1-\alpha)\sum_{i:f_{i,t}=y_t} w_{i,t-1} \\
&\leq \alpha\Phi_t + (1-\alpha)\frac{\Phi_t}{2} \quad \text{(total weight of correct experts is at most half of total weight)} \\
&= \left(\frac{1+\alpha}{2}\right)\Phi_t
\end{aligned}$$

Now, we have $\Phi_{t+1} \leq \Phi_t, \forall t$.

Let M_t be the number of mistakes made by the algorithm up to time t and T be the stopping time. We have,

$$\Phi_{T+1} \leq \left(\frac{1+\alpha}{2}\right)^{M_T}\Phi_1 = \left(\frac{1+\alpha}{2}\right)^{M_T} N$$

Let $M_{i,t}$ TODO: complete this

2 August 4, 2025 (Class 2)

3 August 7, 2025 (Class 3)

3.0.1 Convex function

A function $f:\mathbb{R}^d \to \mathbb{R}$ is convex if for all $x_1, x_2 \in \mathbb{R}^d$ and $\lambda \in [0,1]$,

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

3.0.2 Strictly convex function

A function $f:\mathbb{R}^d \to \mathbb{R}$ is strictly convex if for all $x_1, x_2 \in \mathbb{R}^d$, $x_1 \neq x_2$ and $\lambda \in (0,1)$,

$$f(\lambda x_1 + (1-\lambda)x_2) < \lambda f(x_1) + (1-\lambda)f(x_2)$$

3.0.3 strongly convex function

A function $f:\mathbb{R}^d \to \mathbb{R}$ is α-strongly convex if for all $x_1, x_2 \in \mathbb{R}^d$ and $\lambda \in [0,1]$,

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2) - \frac{\alpha}{2}\lambda(1-\lambda)||x_1 - x_2||^2$$

3.0.4 Equivalent conditions of strongly convex function

Let f be a differentiable function on a convex set X. The following statements are equivalent:

- f is α-strongly convex on X.
- For all $x_1, x_2 \in X$, $\lambda \in [0,1]$,
$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2) - \frac{\alpha}{2}\lambda(1-\lambda)||x_1 - x_2||^2$$
- For all $x_1, x_2 \in X$,
$$f(x_1) \geq f(x_2) + \nabla f(x_2)^T(x_1 - x_2) + \frac{\alpha}{2}||x_1 - x_2||^2$$
- For all $x_1, x_2 \in X$,
$$(\nabla f(x_1) - \nabla f(x_2))^T(x_1 - x_2) \geq \alpha||x_1 - x_2||^2$$
- If f is twice differentiable, then for all $x \in X$,
$$\nabla^2 f(x) \succeq \alpha I$$
- $f(x) - \frac{\alpha}{2}\|x\|^2$ is convex.

3.0.5 smooth function

A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is β-smooth if for all $x_1, x_2 \in \mathbb{R}^d$,

$$f(x_1) \leq f(x_2) + \nabla f(x_2)^T(x_1 - x_2) + \frac{\beta}{2}||x_1 - x_2||^2$$

3.0.6 exp concave function

A function $f : \mathbb{R}^d \to \mathbb{R}$ is α-exp concave if for all $x_1, x_2 \in \mathbb{R}^d$ and $\lambda \in [0,1]$,

$$h(x) = e^{-\alpha f(x)}$$

is concave.

3.0.7 Jensen's inequality

Let ϕ be a convex function and X be a random variable. Then,

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$$

3.0.8 Projection operator

The projection operator $\Pi_{\mathcal{S}}(y)$ is defined as

$$\Pi_{\mathcal{S}}(y) = \arg\min_{x \in \mathcal{S}} ||x - y||$$

4 August 11, 2025 (Class 4)

5 August 14, 2025 (Class 5)

5.1 Exponential Weights Algorithm (action setting)

Algorithm 2 Exponential Weights Algorithm (action setting)

Input: N experts, learning rate $\eta > 0$, convex loss function l
Initialize: $w_{i,1} = 1 \quad \forall i \in [N]$
for $t = 1, 2, \ldots$ **do**
    Input: expert predictions $f_{i,t} \in \mathcal{D}$ (convex set)
    Algorithm takes action: $a_t = \frac{\sum_{i=1}^{N} w_{i,t-1} f_{i,t}}{\sum_{i=1}^{N} w_{i,t-1}}$
    Adversary selects: $y_t \in \mathcal{Y}$
    Algorithm incurs loss: $l(a_t, y_t)$
    Update weights: $w_{i,t} \leftarrow w_{i,t-1} \cdot e^{-\eta\, l(f_{i,t}, y_t)}$
end for

We define the best action in hindsight as

$$a^* = \arg\min_{i \in [N]} \sum_{t=1}^{T} l(f_{i,t}, y_t)$$

Similarly, we define the overall regret as

$$R_T(\text{exp wts}) = \mathbb{E} \sum_{t=1}^{T} l(a_t, y_t) - \underbrace{\sum_{t=1}^{T} l(a^*, y_t)}_{\text{benchmark}}$$

5.1.1 An optimal regret bound for Exponential Weights Algorithm for general convex loss functions

For a general convex loss function l bounded in $[a, b]$, the Exponential Weights Algorithm with learning rate $\eta > 0$ achieves the following regret bound against any sequence of outcomes $y_1, \ldots, y_T$. The bound is given by,

$$R_T(\text{exp wts}) \le \frac{\ln(N)}{\eta} + \frac{\eta(b-a)^2 T}{8}.$$

In particular, choosing $\eta = \sqrt{\frac{8 \ln(N)}{(b-a)^2 T}}$, we have,

$$R_T(\text{exp wts}) \le (b-a)\sqrt{\frac{T}{2} \ln(N)}$$

Proof:

Preliminaries:

We assume that the loss function l is convex and bounded in $[a, b]$.

We define,

$$\begin{aligned} L_{i,T} &:= \sum_{t=1}^{T} l(f_{i,t}, y_t) \\ \hat{L}_T &:= \sum_{t=1}^{T} l(a_t, y_t) \\ w_{i,t} &:= e^{-\eta L_{i,t}} = e^{-\eta \sum_{s=1}^{t} l(f_{i,s}, y_s)}, \quad \forall t \geq 1 \\ W_t &:= \sum_{i=1}^{N} w_{i,t} = \sum_{i=1}^{N} e^{-\eta L_{i,t}}, \quad \forall t \geq 1 \end{aligned}$$

with $W_0 = N$.

We have the following form of the Hoeffding's lemma. Let X be a random variabe with $a \leq X \leq b$ almost surely, then for any $s \in \mathbb{R}$,

$$\ln \mathbb{E}[e^{sX}] \leq s\mathbb{E}[X] + \frac{s^2(b-a)^2}{8}$$

Main proof:

We have,

$$\begin{aligned} \ln \frac{W_n}{W_0} &= \ln \left(\sum_{i=1}^{N} \exp\left(-\eta L_{i,n}\right) \right) - \ln N \\ &\geq \ln \left(\max_{i \in [N]} e^{-\eta L_{i,n}} \right) - \ln N \\ &= -\eta \min_{i \in [N]} L_{i,n} - \ln N \end{aligned}$$

For $t = 1, \ldots, T$,

$$\begin{aligned} \ln \frac{W_t}{W_{t-1}} &= \ln \frac{\sum_{i=1}^{N} e^{-\eta l(f_{i,t}, y_t)} e^{-\eta L_{i,t-1}}}{\sum_{i=1}^{N} e^{-\eta L_{i,t-1}}} \\ &= \ln \frac{\sum_{i=1}^{N} w_{i,t-1} e^{-\eta l(f_{i,t}, y_t)}}{\sum_{i=1}^{N} w_{i,t-1}} \end{aligned}$$

Using Hoeffding's lemma, we have,

$$\begin{aligned} \ln \frac{\sum_{i=1}^{N} w_{i,t-1} e^{-\eta l(f_{i,t}, y_t)}}{\sum_{i=1}^{N} w_{i,t-1}} &\leq -\eta \sum_{i=1}^{N} \left(\frac{w_{i,t-1}}{\sum_{i=1}^{N} w_{i,t-1}} \right) l(f_{i,t}, y_t) + \frac{\eta^2 (b-a)^2}{8} \\ &= -\eta l(a_t, y_t) + \frac{\eta^2 (b-a)^2}{8} \quad \text{(by convexity of } l) \\ \implies \ln \frac{W_t}{W_{t-1}} &\leq -\eta l(a_t, y_t) + \frac{\eta^2 (b-a)^2}{8} \end{aligned}$$

Now, summing over $t = 1, \dots, T$, we have,

$$\begin{aligned}
\ln \frac{W_T}{W_0} &= \sum_{t=1}^{T} \ln \frac{W_t}{W_{t-1}} \\
&\le -\eta \sum_{t=1}^{T} l(a_t, y_t) + \frac{\eta^2 (b-a)^2 T}{8} \\
\implies \ln \frac{W_T}{W_0} &\le -\eta \hat{L}_T + \frac{\eta^2 (b-a)^2 T}{8}
\end{aligned}$$

Using the lower bound on $\ln \frac{W_T}{W_0}$, we derived earlier, we have,

$$\hat{L}_T - \min_{i \in [N]} L_{i,T} \le \frac{\ln(N)}{\eta} + \frac{\eta (b-a)^2 T}{8}$$

Choosing the optimal $\eta = \sqrt{\frac{8 \ln(N)}{(b-a)^2 T}}$, we have,

$$R_T(\text{exp wts}) \le (b-a) \sqrt{\frac{T}{2} \ln(N)}$$

6 August 21, 2025 (Class 6)

Consider $\mathcal{D} = [0, 1]$, squared loss f^n and 3 experts f_1, f_2, f_3 with constant predictions $f_1(t) = 0$, $f_2(t) = 0.5$, $f_3(t) = 1$, and $\mathcal{Y} = \{0, 1\}$.

6.1 An optimal regret bound for Exponential Weights Algorithm for exp concave loss functions

Consider $\eta - exp$ concave loss function, then the regret is bounded by

$$R_T(\text{exp wts}) \le \frac{\log(N)}{\eta}$$

Proof: We define,

$$\begin{aligned}
L_{i,T} &:= \sum_{t=1}^{T} l(f_{i,t}, y_t) \\
\hat{L}_T &:= \mathbb{E} \sum_{t=1}^{T} l(a_t, y_t)
\end{aligned}$$

We have,

$$
\begin{aligned}
R_T(\text{exp wts}) &= \mathbb{E}\sum_{t=1}^{T} l(a_t, y_t) - \min_{i\in[N]} \sum_{t=1}^{T} l(f_{i,t}, y_t) \\
&= \max_{i\in[N]} \left[\mathbb{E}\sum_{t=1}^{T} l(a_t, y_t) - \sum_{t=1}^{T} l(f_{i,t}, y_t)\right] \\
&= \max_{i\in[N]} \left[\hat{L}_T - L_{i,T}\right] \\
&= \log\left[\max_{i\in[N]} e^{\left(\hat{L}_T - L_{i,T}\right)}\right] \\
&= \frac{1}{\eta}\log\left[\max_{i\in[N]} e^{\left(\eta\left(\hat{L}_T - L_{i,T}\right)\right)}\right] \\
&= \frac{1}{\eta}\log\left[e^{\left(\eta\left(\hat{L}_T - L_{i^*,t}\right)\right)}\right] \qquad \text{where } i^* = \arg\max_{i\in[N]} e^{\left(\hat{L}_T - L_{i,T}\right)} \\
&\leq \frac{1}{\eta}\log\left[\sum_{i=1}^{N} e^{\left(\eta\left(\hat{L}_T - L_{i,T}\right)\right)}\right] \\
\Phi(T) &= \frac{1}{\eta}\log\left[\sum_{i=1}^{N} e^{\left(\eta\left(\hat{L}_T - L_{i,T}\right)\right)}\right]
\end{aligned}
$$

Now, we have,

$$
\begin{aligned}
R_T(\text{exp wts}) &\leq \Phi(T) \\
\text{where } \Phi(T) &= \frac{1}{\eta}\log\left[\sum_{i=1}^{N} e^{\left(\eta\left(\hat{L}_T - L_{i,T}\right)\right)}\right] \\
\text{and } \Phi(0) &= \frac{\log(N)}{\eta}
\end{aligned}
$$

It is sufficient to show $\Phi(t) \leq \Phi(t-1)\ \forall t$.

$$
\begin{aligned}
\sum_{i=1}^{N} e^{\left(\eta\left(\hat{L}_t - L_{i,t}\right)\right)} &\leq \sum_{i=1}^{N} e^{\left(\eta\left(\hat{L}_{t-1} - L_{i,t-1}\right)\right)} \\
\sum_{i=1}^{N} e^{(\eta\hat{L}_{t-1}) + (\eta\hat{l}_t) - (\eta L_{i,t-1}) - (\eta l_{i,t})} &\leq \sum_{i=1}^{N} e^{(\eta\hat{L}_{t-1}) - (\eta L_{i,t-1})} \\
\implies \sum_{i=1}^{N}\left(\frac{e^{-\eta L_{i,t-1}}}{\sum_{i=1}^{N} e^{-\eta L_{o,t-1}}}\right) e^{-\eta l_{i,t}} &\leq e^{\eta\hat{l}_t}
\end{aligned}
$$

TODO: Complete this.

7 August 25, 2025 (Class 7)

7.1 Stock Portfolio Optimization

We have N stock options, and δ_N is the set of all distribution over $[N]$. We take initial wealth as 1, without loss of generality.

We define,

$$r_{i,t} = \frac{\text{closing value of stock i on day t}}{\text{opening value of stock i on day t}}$$

$$(P_t)_{t \leq T} : \text{Action sequence, the distribution of wealth on day t}$$
$$(r_t)_{t \leq T} : \text{Market (reward) sequence, the return of each stock on day t}$$

We have the wealth at stopping day T,

$$W_T = \prod_{t=1}^{T} \langle P_t, r_t \rangle$$

We have the max possible wealth as,

$$W_T^* = \max_{P \in \delta_N} \prod_{t=1}^{T} \langle P_t, r_t \rangle$$

and the best constant distribution as,

$$P^* = \arg \max_{P \in \delta_N} \prod_{t=1}^{T} \langle P_t, r_t \rangle$$

$$\log \left(\frac{W_T}{W_T^*} \right) = \sum_{i=1}^{T} \log(\langle P_t, r_t \rangle) - \sum_{i=1}^{T} \log(\langle P^*, r_t \rangle)$$
$$l(P_t, r_t) = -\log(\langle P_t, r_t \rangle `1')$$

8 September 1, 2025 (Class 8)

9 September 4, 2025 (Class 9)

We have an online convex optimization setting

Let $\mathcal{K}$ be the set of values x_t which the algorith can choose, and $\mathcal{F}$ be the set of all convex loss functions.

Algorithm: Online Convex Optimization
input: $\mathcal{K} :=$ set of values, $\mathcal{F} :=$ set of convex loss functions
for $t = 1, 2, \ldots$ **do**
– Algorithm selects action $x_t \in \mathcal{K}$
– Environment selects a loss function $f_t \in \mathcal{F}$
– ALG incurs loss $f_t(x_t)$
end for

Do note, the env only gives $f_t(x_t)$ and $\nabla f_t(x_t)$, not f_t itself.

9.1 Online gradient descent

Algorithm: Online Gradient Descent
input: $\mathcal{K} :=$ set of values, $\mathcal{F} :=$ set of convex loss functions
for $t = 1, 2, \ldots$ **do**
– Algorithm selects action $x_t \in \mathcal{K}$
– Environment selects a loss function $f_t \in \mathcal{F}$
– ALG incurs loss $f_t(x_t)$
– Update $x_{t+1} \leftarrow \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t))$
end for

9.1.1 Regret guarantee on online gradient descent

We have,

$$R_T(OGD) \leq \frac{3}{2} DG\sqrt{T},$$

where

$$G := \sup_{f \in \mathcal{F}} \sup_{x \in \mathcal{K}} \|\nabla f(x)\|$$
$$D := \sup_{x, x' \in \mathcal{K}} \|x - x'\|_2$$

Proof: We define

$$\nabla_t := \nabla f_t(x)$$

Also, $x^* \in \mathcal{K}$ as

$$x^* = \arg\min_{x \in \mathcal{K}} \sum_{i=1}^{T} f_i(x).$$

From the first order condition of convexity,

$$\begin{aligned}
f_t(x_t) - f_t(x^*) &\leq \nabla_t^T (x_t - x^*) \\
\sum_{t=1}^{T} f_t(x_t) - \sum_{t=1}^{T} f_t(x^*) &\leq \sum_{t=1}^{T} \nabla_t^T (x_t - x^*) \\
\implies R_T(OGD) &\leq \sum_{t=1}^{T} \nabla_t^T (x_t - x^*).
\end{aligned}$$

Now,

$$\begin{aligned}
\|x^* - x_{t+1}\| &= \|x^* - \Pi_{\mathcal{K}}(x_t - \eta_t \nabla_t)\| \\
&\leq \|x^* - x^t + \eta_t \nabla_t\|, \quad \text{by Pythogorean theorem} \\
\|x^* - x_{t+1}\|^2 &\leq \|x^* - x_t\|^2 + \eta_t^2 \|\nabla_t\|^2 + 2\eta_t \langle x^* - x_t, \nabla_t \rangle \\
-2\eta_t \langle x^* - x_t, \nabla_t \rangle &\leq \|x^* - x_t\|^2 - \|x^* - x_{t+1}\|^2 + \eta_t^2 \|\nabla_t\|^2 \\
2 \langle x_t - x^*, \nabla_t \rangle &\leq \frac{1}{\eta_t} \left(\|x^* - x_t\|^2 - \|x^* - x_{t+1}\|^2 \right) + \eta_t \|\nabla_t\|^2
\end{aligned}$$

From the above two results, we have

$$
\begin{aligned}
2R_T(OGD) &\leq \sum_{t=1}^{T}\left(\frac{\|x^* - x_t\|^2 - \|x^* - x_{t+1}\|^2}{\eta_t}\right) + \sum_{t=1}^{T}\eta_t\|\nabla_t\|^2 \\
&= \frac{\|x^* - x_1\|^2}{\eta_1} + \sum_{t=2}^{T}\|x^* - x_t\|^2\left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}}\right) - \frac{\|x^* - x_{T+1}\|^2}{\eta_T} + \sum_{t=1}^{T}\eta_t\|\nabla_t\|^2 \\
&\leq \frac{\|x^* - x_1\|^2}{\eta_1} - \frac{D^2}{\eta_1} + \frac{D^2}{\eta_T} - \frac{\|x^* - x_{T+1}\|^2}{\eta_T} + G^2\sum_{t=1}^{T}\eta_t \\
&\leq \frac{D^2}{\eta_T} + G^2\sum_{t=1}^{T}\eta_t
\end{aligned}
$$

Putting $\eta_t = \frac{D}{G\sqrt{t}}$, we have

$$
\begin{aligned}
2R_T(OGD) &\leq D^2\frac{G\sqrt{T}}{D} + G^2\frac{D}{G}\sum_{t=1}^{T}\frac{1}{\sqrt{t}} \\
&\leq DG\sqrt{T} + 2DG\sqrt{T} \\
\implies R_T(OGD) &\leq \frac{3}{2}DG\sqrt{T}
\end{aligned}
$$

9.1.2 Regret guarantee on online gradient descent for strongly convex loss function

With $\eta = 1/(\alpha t)$, we have

$$R_T(OGD) \leq \frac{G^2}{2\alpha}(1 + \log(T))$$

Proof: From α-strong convexity property we have

$$2(f_t(x_t) - f_t(x^*)) \leq 2\nabla_t^T(x_t - x^*) - \alpha\|x_t - x^*\|^2$$

We will use the upper bound on the first term from previous proof. We have TODO: check and correct this.

$$
\begin{aligned}
2R_T(\text{OGD}) &= \sum_{t=1}^{T} 2(f_t(x_t) - f_t(x^*)) \\
&\leq \sum_{t=1}^{T}\left(\frac{\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2}{\eta_t} - \alpha\|x_t - x^*\|^2\right) + G^2\sum_{t=1}^{T}\eta_t - \frac{\|x_{T+1} - x^*\|^2}{\eta_T} \\
&\leq D^2\underbrace{\sum_{t=1}^{T}\left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} - \alpha\right)}_{=0} + G^2\sum_{t=1}^{T}\eta_t \\
&= \frac{G^2}{\alpha}\sum_{t=1}^{T}\frac{1}{t} \\
&\leq \frac{G^2}{\alpha}(1 + \log(T))
\end{aligned}
$$

9.2 Follow the leader (FTL)

Algorithm: Follow The Leader (FTL)
Input: A convex set of possible actions (decision set) $\mathcal{K}$.

for $t = 1, 2, 3, \ldots, T$ **do**

1. Play the action x_t that minimizes the cumulative loss observed so far:
   $x_t := \arg\min_{x \in \mathcal{K}} \sum_{s=1}^{t-1} f_s(x)$
   (Note: For the first round, $t = 1$, the sum is empty, so any action $x_1 \in \mathcal{K}$ can be chosen.)

2. The environment reveals the loss function for the current round, f_t.

3. The algorithm incurs and observes the loss $f_t(x_t)$.

end for

9.2.1 FTL-BTL lemma

Lemma 9.1 (FTL-BTL lemma[1]): *Let $x_1, x_2, \ldots$ be the sequence of points chosen by FTL. Then, for any $u \in \mathcal{K}$ and for and stopping time T, we have*

$$\sum_{t=1}^{T} f_t(x_{t+1}) \leq \sum_{t=1}^{T} f_t(u)$$

Proof: TODO: using induction
TODO: Follow the Regularized Leader For linear loss

9.3 Follow the regularized leader (FTRL)

Remark 9.2: *The regularizers considered are bounded, α-strongly convex functions.*

Algorithm 11: FTRL Algorithm for general loss function
Input: convex set $\mathcal{K}$, function class $\mathcal{F}$, regularization function R
Initialize: $x_1 = \arg\min_{x \in \mathcal{K}} R(x)$
for $t = 1, 2, \ldots$ **do**
– Algorithm plays $x_t \in \mathcal{K}$
– Environment reveals $f_t \in \mathcal{F}$
– Algorithm incurs a loss $f_t(w_t) \in \mathbb{R}$
– Update
$$x_{t+1} := \arg\min_{x \in \mathcal{K}} \left[\sum_{s=1}^{t} \langle \nabla_s, x \rangle + \frac{1}{\eta} R(x) \right]$$
end for

[1] Follow the leader, be the leader lemma

9.3.1 FTRL with linear loss function and negative entropic regularizer

10 September 8, 2025 (Class 10)

10.1 Some mathematical preliminaries

Definition 10.1 (Bregman divergence): *We define the Bregman divergence, with respect to function R, as*

$$B_R(x\|y) = R(x) - R(Y) - \langle \nabla R(x), x - y \rangle$$

Remark 10.2: *For twice differentiable functions, the Bregman divergence is equal to the second derivative at an intermediate point.*

note: For $R(x) = \frac{1}{2}\|x\|_{L_2}^2$, we have

$$B_R(x\|y) = \frac{1}{2}\|x - y\|^2.$$

For $R(x) = \sum_{i=1}^d x_i \log x_i$, $\sum x_i = 1$ and $d = 2$, we have

$$B_R(x\|y) = D_{KL}(x\|y)$$

Properties of Bregman divergence:

- It is strictly convex in the first argument
- Non-negative $B_R(x\|y) \geq 0$. The proof follows from Taylors theorem.
- Asymmetric i.e. $B_R(x\|y) \neq B_R(y\|x)$.
- Non-convex in the second argument. Example $R(x) = -\log(x)$ and $(B_R(x\|y) = \log(y) - \log(x) - \frac{x-y}{y})$
- $\frac{\partial B_R(x\|y)}{\partial x} = \nabla R(x) - \nabla R(y)$.
- Cosine inequality

$$B_R(x\|y) + B_R(y\|z) = B_R(x\|z) + \langle x - y, \nabla R(z) - \nabla R(y) \rangle$$

Definition 10.3 (Dual norm): *Let $\|\cdot\|_*$ be a norm on the vector space $\mathcal{K} \subseteq \mathbb{R}^n$, then the function $\|\cdot\|^*$ defined as*

$$\|x\|^* = \sup_{y \in \mathcal{K}, \|y\|_* \leq 1} \langle x, y \rangle = \sup_{y \in \mathcal{K}} \left\langle x, \frac{y}{\|y\|_*} \right\rangle,$$

is called a dual norm.

Definition 10.4 (Norm induced by a symmetric positive definite matrix): *Let $A \in S_{++}^n$ [2] be a matrix. We define the norm induced by A as*

$$\|x\|_A = \sqrt{x^T A x}$$

[2] symmetric positive definite matrix

Lemma 10.5: *Let $\|x\|_A$ be a norm where $A \in S^n_{++}$. We have the dual norm*

$$\|x\|_A^* = \sqrt{x^T A^{-1} x}$$

Proof:

11 September 11, 2025 (Class 11)

11.1 FTRL for general loss functions

Algorithm: FTRL Algorithm for general loss function
Input: convex set $\mathcal{K}$, function class $\mathcal{F}$, regularization function R
Initialize: $x_1 = \arg\min_{x \in \mathcal{K}} R(x)$
for $t = 1, 2, \ldots$ **do**
– Algorithm plays $x_t \in \mathcal{K}$
– Environment reveals $f_t \in \mathcal{F}$
– Algorithm incurs a loss $f_t(w_t) \in \mathbb{R}$
– Update
$x_{t+1} := \arg\min_{x \in \mathcal{K}} \left[\sum_{s=1}^t \langle \nabla_s, x \rangle + \frac{1}{\eta} R(x)\right]$
end for

Lemma 11.1: *Let $x_1, x_2, \ldots$ be a sequence of points chosen by FTRL, then for any $u \in \mathcal{K}$ and $t \geq 1$, we have*

$$\sum_{s=1}^t f_s(x_{s+1}) - \sum_{s=1}^t f_s(u) \leq \frac{R(u) - R(x_1)}{\eta}$$

:

11.1.1 Regret upper bound of FTRL

We have the regret upper bound as

$$R_T(FTRL) \leq \eta \sum_{t=1}^T (\|\nabla_t\|_t^*)^2 + \frac{R(u) - R(x_1)}{\eta}$$

Proof: TODO

12 September 15, 2025 (Class 12)

12.1 Online Mirror Descent (OMD)

Algorithm: Online Mirror Descent (OMD)
Input: convex set $\mathcal{K}$, function class $\mathcal{F}$, regularization function R
Initialize: $x_1 = \arg\min_{x \in \mathcal{K}} R(x)$ and y_1 such that $\nabla R(y_1) = 0$;
for $t = 1, 2, \ldots$ **do**
- Algorithm plays $x_t \in \mathcal{K}$;
- Environment reveals $f_t \in \mathcal{F}$;
- Algorithm incurs a loss $f_t(x_t) \in \mathbb{R}$;
- Update
  y_{t+1} such that $\nabla R(y_{t+1}) = \nabla R(y_t) - \eta \nabla_t$ [Lazy version]
  y_{t+1} such that $\nabla R(y_{t+1}) = \nabla R(x_t) - \eta \nabla_t$ [Agile version]

  - $x_{t+1} = \arg\min_{x \in \mathcal{K}} B_R(x \| y_{t+1})$

end

12.1.1 Equivalence of lazy OMD and FTRL

Theorem 12.1 (Equivalence of lazy OMD and FTRL): *Let $\mathcal{F}$ be the class of loss functions. Then the lazy OMD and FTRL algorithms play the same points,*

$$\arg\min_{x \in \mathcal{K}} (B_R(x \| y)) = \arg\min_{x \in \mathcal{K}} \left(\eta \sum_{s=1}^{t-1} \nabla_s^T x + R(x) \right)$$

Proof: TODO

12.1.2 Agile OMD

Theorem 12.2: *Let $\mathcal{K}$ be a convex set and $x' = \Pi_{\mathcal{K}} B_R(x \| y)$ be a Bregman projection of some $y \in \mathbb{R}^n$ on $\mathcal{K}$ and $u \in \mathcal{K}$. Then*

$$B_R(y \| u) \geq B_R(y \| x) + B_R(x \| u)$$

Theorem 12.3 (Regret bound of Agile OMD): *For $\forall u \in \mathcal{K}$, we have*

$$R_T(OMD) \leq \frac{\eta}{2} \sum_{t=1}^{T} (\|\nabla\|_t^*)^2 + \frac{D_R^2}{\eta}$$

13 September 22, 2025 (Class 13)

13.1 Bandits setting

Algorithm: Multi-Armed Bandits setting
input: $n :=$ number of arms
for $t = 1, 2, \ldots$ **do**
  – Algorithm picks an arm $i_t \in [n]$
  – Environment reveals loss $l_{i_t,t} \in [0, 1]$
end for

13.2 Naive application of online gradient descent for bandit setting

Algorithm: Online gradient descent for bandit setting
input: $n :=$ number of arms, $\delta \in (0, 1)$
for $t = 1, 2, \ldots$ **do**
  $b_t \sim \text{Bern}(\delta)$
  if $b_t = 1$ **then**
    - $i_t \sim \text{Unif}([n])$ Exploration
    - $\hat{\ell}_{i,t} = \begin{cases} \frac{n}{\delta}\ell_{i,t} & \text{if } i = i_t \\ 0 & \text{otherwise} \end{cases}$
    - Set $\hat{f}_t = \hat{\ell}_{i,t} \cdot x_{i,t}$ for all $i \in [n]$, i.e. $\hat{f}_t = \langle \hat{\ell}_t, x \rangle$
    - $x_{t+1} = \text{OGD}(\hat{f}_1, \hat{f}_2, \ldots, \hat{f}_t)$
  end if
  else if $b_t = 0$ **then**
    - $i_t \sim x_t$ Exploitation
    - $x_{t+1} = x_t, \hat{\ell}_t = 0$ and $\hat{f}_t = 0$
  end if
end for

Lemma 13.1: $\mathbb{E}[l_{i_t,t}] \leq \mathbb{E}[\langle \hat{l}_t, x_t \rangle] + \delta$

Proof:

13.2.1 Regret upper bound of OGD-MAB

We have,

$$\mathbb{E}\left[R_T(OGD - MAB)\right] \leq 4(nT)^{2/3},$$

where we chose $\delta = n^{2/3}T^{-1/3}$

Proof:

$$\begin{aligned}
\mathbb{E}[R_T(OGD-MAB)] &= \mathbb{E}\left[\sum_{t=1}^T l_{i_t,t} - \sum_{t=1}^T l_{i^*,t}\right] \\
&= \mathbb{E}\left[\sum_{t=1}^T l_{i_t,t} - \sum_{t=1}^T \hat{l}_{i^*,t}\right] \\
&\leq \mathbb{E}\left[\sum_{t=1}^T \langle l_t, x_t\rangle - \sum_{t=1}^T \hat{l}_{i^*,t}\right] + \delta T \\
&= \mathbb{E}[R_T(OGD)] + \delta T \\
&\leq \frac{3}{2}GD\sqrt{\delta T} + \delta T, \qquad G = \sup\|\hat{l}\| \leq n/\delta, D \leq 2 \\
&\leq \frac{3n}{\delta}\sqrt{\delta T} + \delta T \\
&\leq \frac{3n}{\delta}\sqrt{n}
\end{aligned}$$

14 September 25, 2025 (Class 14)

14.1 EXP3-γ

Algorithm : EXP3-γ
Input: Number of arms $n, \gamma \in (0,1)$
Initialize: $w_{i,1} = 1$ for all $i \in [n]$
for $t = 1, 2, \ldots$ **do**
- $p_{i,t} = (1-\gamma)\frac{w_{i,t}}{\sum_{j=1}^n w_{j,t}} + \frac{\gamma}{n}$
- Draw $i_t \sim p_t$ where $p_t = \begin{pmatrix} p_{1,t} & p_{2,t} & \cdots & p_{n,t}\end{pmatrix}^T$
- Observe $r_{i_t,t}$, the reward at time t (from arm i_t)
- $\hat{r}_{i,t} = \mathbb{I}\{i_t = i\}\dfrac{r_{i_t,t}}{p_{i_t,t}}$
- $w_{i,t+1} = w_{i,t} \cdot e^{\eta \hat{r}_{i,t}}$

end

Theorem 14.1: *For any $\gamma \in (0,1)$, any reward sequence $(r_t)_{t\geq 1}$ with $r_t \in [0,1]^n$, we have*

$$\mathbb{E}[R_T(EXP3-\gamma)] = \max_i \sum_{t=1}^T r_{i,t} - \mathbb{E}\left[\sum_{t=1}^T r_{i_t,t}\right] \leq 2\gamma T + \frac{n\log(n)}{\gamma}.$$

Furthermore, taking $\gamma = \sqrt{\dfrac{n\log(n)}{2T}}$,

$$\mathbb{E}[R_T(EXP3-\gamma)] \leq 2\sqrt{2Tn\log(n)}$$

Proof: Let $W_t = \sum_{i=1}^n w_{i,t}$.

Upper bounding $\log(W_{T+1}/W_1)$,

$$\begin{aligned}
\frac{W_{t+1}}{W_t} &= \sum_{i=1}^{n} \frac{w_{i,t+1}}{W_t} = \sum_{i=1}^{n} \left(\frac{w_{i,t}}{W_t}\right) e^{\gamma \hat{r}_{i,t}/n} \\
&= \sum_{i=1}^{n} \frac{p_{i,t} - \gamma/n}{1-\gamma} \cdot \exp\left(\frac{\gamma \hat{r}_{i,t}}{n}\right) \\
&\leq \sum_{i=1}^{n} \frac{p_{i,t} - \gamma/n}{1-\gamma} \cdot \exp\left(\frac{\gamma \hat{r}_{i,t}}{n}\right) \\
&\leq \sum_{i=1}^{n} \frac{p_{i,t} - \gamma/n}{1-\gamma} \left(1 + \frac{\gamma}{n}\hat{r}_{i,t} + \left(\frac{\gamma}{n}\right)^2 \hat{r}_{i,t}^2\right) \\
&\qquad\qquad\qquad\qquad (\text{Using } e^x \leq 1 + x + x^2 \text{ for all } x \in [0,1]) \\
&\leq \frac{1}{1-\gamma} \sum_{i=1}^{n} p_{i,t} \left(1 + \frac{\gamma}{n}\hat{r}_{i,t} + \left(\frac{\gamma}{n}\right)^2 \hat{r}_{i,t}^2\right) \\
&\leq \frac{1}{1-\gamma} \left(\underbrace{\sum_{i=1}^{n} p_{i,t}}_{=1} + \frac{\gamma}{n} \underbrace{\sum_{i=1}^{n} p_{i,t}\hat{r}_{i,t}}_{=r_{i_t,t}} + \left(\frac{\gamma}{n}\right)^2 \sum_{i=1}^{n} \underbrace{p_{i,t}\hat{r}_{i,t}}_{\leq 1} \hat{r}_{i,t} \right) \\
\log\left(\frac{W_{t+1}}{W_t}\right) &\leq \frac{1}{1-\gamma} \left(\frac{\gamma}{n} r_{i_t,t} + \left(\frac{\gamma}{n}\right)^2 \sum_{i=1}^{n} \hat{r}_{i,t}\right), \quad \text{using } \log(1+x) \leq x \text{ and } \frac{1}{1-\gamma} \geq 1 \\
\implies \log\left(\frac{W_{T+1}}{W_1}\right) &\leq \frac{1}{1-\gamma} \left(\frac{\gamma}{n} \sum_{t=1}^{T} r_{i_t,t} + \left(\frac{\gamma}{n}\right)^2 \sum_{t=1}^{T} \sum_{i=1}^{n} \hat{r}_{i,t}\right), \quad \text{summing from } t = 1 \text{ to } T
\end{aligned}$$

Lower bounding $\log(W_{T+1}/W_1)$,

$$\begin{aligned}
\log\left(\frac{W_{T+1}}{W_1}\right) &\geq \log\left(\frac{w_{i,T+1}}{W_1}\right) \\
&= \log(e^{\gamma/n \sum_{t=1}^{T} \hat{r}_{i,t}}) - \log(n), \quad \because W_1 = n \\
&= \frac{\gamma}{n} \sum_{t=1}^{T} \hat{r}_{i,t} - \log(n)
\end{aligned}$$

Let i^* be the best arm in hindsight. Combining the above upper and lower bounds for i^*,

$$
\begin{aligned}
\frac{\gamma}{n}\sum_{t=1}^{T}\hat{r}_{i^*,t}-\log(n) &\leq \frac{1}{1-\gamma}\left(\frac{\gamma}{n}\sum_{t=1}^{T}r_{i_t,t}+\left(\frac{\gamma}{n}\right)^2\sum_{t=1}^{T}\sum_{i=1}^{n}\hat{r}_{i,t}\right)\\
\frac{\gamma}{n}\sum_{t=1}^{T}r_{i^*,t}-\log(n) &\leq \frac{1}{1-\gamma}\left(\frac{\gamma}{n}\mathbb{E}\left[\sum_{t=1}^{T}r_{i_t,t}\right]+\left(\frac{\gamma}{n}\right)^2\sum_{t=1}^{T}\sum_{i=1}^{n}r_{i,t}\right), \quad \because \mathbb{E}[\hat{r}_{i,t}]=r_{i,t}\\
(1-\gamma)\sum_{t=1}^{T}r_{i^*,t}-\mathbb{E}\left[\sum_{t=1}^{T}r_{i_t,t}\right] &\leq \frac{(1-\gamma)n\log(n)}{\gamma}+\left(\frac{\gamma}{n}\right)\sum_{t=1}^{T}\underbrace{\sum_{i=1}^{n}r_{i,t}}_{\leq n}\\
\sum_{t=1}^{T}r_{i^*,t}-\mathbb{E}\left[\sum_{t=1}^{T}r_{i_t,t}\right] &\leq \frac{(1-\gamma)n\log(n)}{\gamma}+\gamma T+\gamma\sum_{t=1}^{T}r_{i^*,t}\\
\mathbb{E}[R_T(EXP3-\gamma)] &\leq \frac{n\log(n)}{\gamma}+2\gamma T, \quad \because r_{i^*,t}\leq 1
\end{aligned}
$$

Taking $\gamma=\sqrt{\dfrac{n\log(n)}{2T}}$,

$$
\mathbb{E}[R_T(EXP3-\gamma)] \leq 2\sqrt{2Tn\log(n)}
$$

Remark 14.2: *Using a tighter bound of e^x in the upper bounding gives a better constant.*

15 October 6, 2025 (Class 15)

15.1 Stochastic Multi Armed Bandits

Lemma 15.1: *Let $N_{i,T}$ be the number if times the arm i has been pulled till time T by an algorithm and let $\Delta_i=\mu_{i^*}-\mu_i$ be the suboptimality gap. Then,*

$$
R_T(ALG)=\sum_{i=1}^{N}\Delta_i\mathbb{E}[N_{i.T}]
$$

Proof:

$$
N_{i,T}=\sum_{t=1}^{T}\mathbb{I}(i_t=i)
$$

We have,

$$\sum_{i=1}^{N} \Delta_i \mathbb{E}[N_{i.T}] = \mathbb{E}\left[\sum_{i=1}^{N} (\mu_{i^*} - \mu_i) \left(\sum_{t=1}^{T} \mathbb{I}(i_t = i)\right)\right]$$
$$= \mathbb{E}\left[\sum_{i=1}^{N} \mu_{i^*} \sum_{t=1}^{T} \mathbb{I}(i_t = i) - \sum_{i=1}^{N} \mu_i \sum_{t=1}^{T} \mathbb{I}(i_t = i)\right]$$
$$= \mu_{i^*} \left(\sum_{t=1}^{T}\sum_{i=1}^{N} \mathbb{E}\left[\mathbb{I}(i_t = i)\right]\right) - \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{N} \mu_i \mathbb{I}(i_t = i)\right]$$
$$= \mu_{i^*} T - \sum_{t=1}^{T} \mu_{i_t}, \qquad \because \sum_{i=1}^{N} \mathbb{E}\left[\mathbb{I}(i_t = i)\right] = 1$$
$$\sum_{i=1}^{N} \Delta_i \mathbb{E}[N_{i.T}] = R_T(ALG)$$

16 October 9, 2025 (Class 16)

16.1 Exploration separate algorithm

We have, from the `[Hoeffding's Inequality]`,

$$Pr\{|\hat{\mu}_{i,t} - \mu_i| \geq \varepsilon_{i,t}\} \leq 2\exp(-2N_{i,t}\varepsilon_{i,t}^2)$$
$$Pr\{\hat{\mu}_{i,t} - \mu_i \geq \varepsilon_{i,t}\} \leq \exp(-2N_{i,t}\varepsilon_{i,t}^2)$$

where

$$N_{i,t} := \text{times arm(i) is pulled till time(t)}$$
$$\hat{\mu}_{i,t} := \text{emperical average till time (t).}$$

We define

$$\delta := \exp(-2N_{i,t}\varepsilon_{i,t}^2)$$
$$\implies \varepsilon_{i,t} = \sqrt{\frac{\log(1/\delta)}{2N_{i,t}}}$$

16.1.1 Regret analysis of exploration separate algorithm

$$R_T(ALG) = \sum_{i=1}^{N} \Delta_i \mathbb{E}[N_{i,t}]$$
$$= \sum_{i=1}^{N} \Delta_i \mathbb{E}[N_{i,t'}] + \sum_{i=1}^{N} \Delta_i \mathbb{E}[N_{i,t'+1:T}]$$

We have,

$$\underbrace{j}_{r.v.} = \arg\max_{i\in[N]} \hat{\mu}_{i,t'}$$

$$i^* = \arg\max_{i\in[N]} \mu_i$$

$$\delta = 1/T^4 \implies \varepsilon_{i,t'} = \sqrt{\frac{2\log T}{N_{i,t'}}}$$

Using the above,

$$\begin{aligned} R_T(ALG) &= \sum_{i=1}^{N} \Delta_i \frac{\alpha T}{N} + (T-t') \sum_{i=1}^{N} \mathbb{I}(i=j)\nabla_j \\ &\leq \alpha T + T\sum_{i=1}^{N} (\mu_{i^*} - \mu_i)\mathbb{I}(i=j), \qquad (\Delta_i < 1; t' > 0) \end{aligned}$$

Now, we have

$$\begin{aligned} \mu_j &\geq \hat{\mu_j} - \varepsilon_j \qquad \text{w.p. atleast } (1 - 1/T^4) \\ \mu_{i^*} - \mu_j &\leq \mu_{i^*} - \hat{\mu}_j + \varepsilon_j \end{aligned}$$

TODO: complete this.

16.1.2 Instance independent (distribution-free) regret

The instance independent (distribution-free) regret bound for the exploration-separate algorithm is

$$R_T(\text{ALG}) \leq \alpha T + T \cdot \mathbb{P}(j \neq i^*)$$

where αT is the regret incurred during the exploration phase, and the second term is the regret incurred during the exploitation phase if the empirically best arm j is not the true best arm i^*.

Using Hoeffding's inequality, with probability at least $1-1/T^4$, the empirical mean of each arm is within $\varepsilon_{i,t'}$ of its true mean. Thus, the probability of choosing a suboptimal arm in the exploitation phase is at most N/T^4, so the regret in the exploitation phase is at most $T \cdot N/T^4 = N/T^3$, which is negligible for large T.

Therefore, the total regret is

$$R_T(\text{ALG}) \leq \alpha T + o(1)$$

16.1.3 Instance dependent (distribution-dependent) regret

The instance dependent regret bound depends on the suboptimality gaps $\Delta_i = \mu_{i^*} - \mu_i$. After the exploration phase, with high probability, the empirically best arm is the true best arm. Thus, the regret is dominated by the exploration phase:

$$R_T(\text{ALG}) \leq \sum_{i:\Delta_i > 0} \Delta_i \cdot \frac{\alpha T}{N} + o(1)$$

where $\frac{\alpha T}{N}$ is the number of times each arm is pulled during exploration.

16.1.4 UCB1 Algorithm

The UCB1 algorithm is an alternative to the exploration-separate algorithm, which balances exploration and exploitation at each round. At time t, for each arm i, define

$$\text{UCB}_i(t) = \hat{\mu}_{i,t} + \sqrt{\frac{2\log t}{N_{i,t}}}$$

where $\hat{\mu}_{i,t}$ is the empirical mean and $N_{i,t}$ is the number of times arm i has been pulled up to time t.

At each round, UCB1 selects the arm with the highest UCB value.

The instance dependent regret of UCB1 is bounded by

$$R_T^{(I.D.)}(\text{UCB1}) \leq \sum_{i:\Delta_i>0} \left(\frac{8\log T}{\Delta_i} + \Delta_i\right)$$

which is logarithmic in T for each suboptimal arm.

17 October 13, 2025 (Midterm 2)

18 October 16, 2025 (Class 17)

19 October 20, 2025 (Class 18)

20 October 23, 2025 (Class 19)

We have the instance independent regret of UCB1 algorithm as

$$R_T^{(I.I.)}(UCB1) = \mathcal{O}\left(\sqrt{NT\log T}\right)$$

Proof of UCB1 instance independent regret bound:

$$event\ E := \left\{ \left| \hat{\mu}_{i,t} - \mu_{i,t} \leq \sqrt{\frac{2\log T}{N_{i,t-1}}} = \varepsilon_{i,t} \quad \forall i, \forall t \leq T \right| \right\}$$
$$\bar{E} := \exists t \leq T, \exists i, \text{such that E above is violated}$$

We have

$$\begin{aligned} Pr(\bar{E}) &\leq \frac{2}{T^4}TN, \quad \text{using union bound} \\ &= \frac{2N}{T^3} \leq \frac{2}{T^2} \quad (N < T) \end{aligned}$$
$$R_T^{(I.I.)}(UCB1) = \mathbb{E}[R_T|E]Pr(E) + \mathbb{E}[R_T|\bar{E}]Pr(\bar{E})$$

Let, without loss of generality, 1 be the best arm.

$$\Delta_{i_t} := \mu_1 - \mu_{i_t} \leq \mu_1 - \mu_{i_t} + \underbrace{\bar{\mu}_{i_t,t}}_{\text{UCB}} - \bar{\mu}_{1,t}$$

$$\mu_1 - \bar{\mu}_{1,t} = \mu_1 - \hat{\mu}_{1,N_{i,t-1}} - \sqrt{\frac{2\log T}{N_{i,t-1}}} \leq 0$$

$$\bar{\mu}_{i_t,t} - \mu_{i_t} = \hat{\mu}_{i_t,N_{i_t,t-1}} + \sqrt{\frac{2\log T}{N_{i_t,t-1}}} - \mu_{i_t} \leq 2\sqrt{\frac{2\log T}{N_{i_t,t-1}}}$$

From the above 3 inequalities,

$$\Delta_{i_t} := \mu_1 - \mu_{i_t} \leq 2\sqrt{\frac{2\log T}{N_{i_t,t-1}}}$$

Now,

$$\begin{aligned}
\sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[N_{i,T}] &= \mathbb{E}\left[\sum_{i:\Delta_i>0} \Delta_i N_{i,T}\right] \\
&\leq \mathbb{E}\left[\sum_{i:\Delta_i>0} 2\sqrt{\frac{2\log T}{N_{i,T}}} N_{i,T}\right] \\
&= \mathbb{E}\left[\sum_{i:\Delta_i>0} \sqrt{8N_{i,T}\log T}\right] \\
&= \sum_{i:\Delta_i>0} \mathbb{E}\left[\sqrt{8N_{i,T}\log T}\right] \\
&= \sqrt{8\log T} N \sum_{i:\Delta_i>0} \frac{1}{N} \mathbb{E}\left[\sqrt{N_{i,t}}\right] \\
&\leq N\sqrt{8\log T}\sqrt{\frac{T}{N}} = \sqrt{8NT\log T} \\
&= \mathcal{O}\left(\sqrt{NT\log T}\right)
\end{aligned}$$

20.0.1 Lower bound on regret for MAB setting

1. Instance independent:

$$\Omega(c\sqrt{NT})$$

2. Instance dependent:

$$\sum_{i:\Delta_i>0} \frac{\Delta_i \log T}{D_{KL}\left(\mu_i \| \mu^*\right)}$$

Some discussion on `[Sub Gaussian Random Variables]`.

20.1 Chernoff bounding technique

Let $\{X_i\}_{i=1}^n \approx$ i.i.d. be mean 0 random variables. Let $\varepsilon, \delta > 0$ and

$$\Psi_X(\lambda) := \log\left(\mathbb{E}\left[\exp(\lambda X)\right]\right).$$

Then, $\exists\, T_0 > 0$ such that

$$Pr\left(\frac{1}{T}\sum_{t=1}^{T} X_t \geq \varepsilon\right) \leq \delta \quad \forall T \geq T_0.$$

Proof of Chernoff bounding technique:

$$\begin{aligned}
Pr\left(\frac{1}{T}\sum_{t=1}^{T} X_t \geq \varepsilon\right) &= Pr\left(\sum_{t=1}^{T} X_t \geq T\varepsilon\right) \\
&= Pr\left(\exp\left(\sum_{t=1}^{T} \lambda X_t\right) \geq \exp(\lambda T\varepsilon)\right) \\
&\leq \mathbb{E}\left[\exp\left(\sum_{t=1}^{T} \lambda X_t\right)\right] \exp(-\lambda T\varepsilon) \\
&= \exp(-\lambda T\varepsilon) \prod_{t=1}^{T} \mathbb{E}\left[\exp\left(\lambda X_t\right)\right] \\
&= \exp(-\lambda T\varepsilon) \prod_{t=1}^{T} \exp\left(\Psi_X(\lambda)\right) \quad (i.i.d.) \\
&= \exp(-\lambda T\varepsilon) \exp\left(T\Psi_X(\lambda)\right) \\
&= \exp\left(-T\left(-\Psi_X(\lambda) + \lambda\varepsilon\right)\right)
\end{aligned}$$

Define $\Psi^*_\lambda(\varepsilon)$ as,

$$\Psi^*_\lambda(\varepsilon) := \sup_{\lambda > 0}\left[\lambda\varepsilon - \Psi_X(\lambda)\right]$$

and so

$$Pr\left(\frac{1}{T}\sum_{t=1}^{T} X_t \geq \varepsilon\right) \leq \exp\left(-T\Psi^*_\lambda(\varepsilon)\right) = \delta$$

21 October 27, 2025 (Class 20)

21.1 Thompson Sampling

α_{i_t} is equal $S_{i_t} + 1$, β_{i_t} is equal $N_{i_t} + 1$, where S and N are success and instances when arm i is pulled.

Algorithm : Thompson Sampling
Input: Number of arms n
Initialize: $(Prior) Beta_i(1,1)$ for all $i \in [n]$
for $t = 1, 2, \dots$ **do**
- Draw $\theta_i \sim Beta_i(\alpha_{i_t}, \beta_{i_t}) \quad \forall i \in [n]$
- Choose arm $i_t = \arg\max_{i \in [n]}(\theta_i)$
- Observe $r_{i_t,t} \sim Bernoulli(\mu_{i_t})$
- Update

$$\alpha_{i_t} \leftarrow \alpha_{i_t} + r_{i_t,t}$$
$$\beta_{i_t} \leftarrow \beta_{i_t} + 1$$

end

TODO: correct the above algorithm.

21.1.1 Regret order of Thompson Sampling

1. Instance dependent regret

$$\mathcal{O}\left(\sum_{i=1}^{n} \frac{\Delta_i(\log T + \log(\log T))}{D_{KL}\left(\mu_i \| \mu_{i^*}\right)}\right)$$

2. Instance independent regret

$$\Omega\left(c\sqrt{NT}\right)$$

21.2 Pure exploration setting

Find the arm which gives the best expected reward.

1. Fixed confidence problem: Given $\varepsilon, \delta > 0$, goal is to find j such that the below probability is satisfied in least number of pulls.

$$Pr(\mu_j \geq \mu_{i^*} - \varepsilon) \geq 1 - \delta$$

2. Fixed budget problem: complementary to the above problem.

21.3 Anytime confidence interval

We have,

$$Pr\left(\bigcup_{t=1}^{\infty} |\hat{\mu}_{i,t} - \mu_i| \geq \sqrt{\frac{\log(4t^2/\delta)}{2t}}\right) \leq \delta$$

Proof: TODO: write proof

We also define

$$U(t,\delta) := \sqrt{\frac{\log(4t^2/\delta)}{2t}}$$

21.4 Successive elimination

21.5 See also

21.6 References